

The General-Rank Square Endpoint for MCG-Finite Surface-Group Representations

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Abstract

Landesman–Litt prove that an MCG-finite representation of a punctured genus- g surface group into $\mathrm{GL}_r(\mathbf{C})$ has finite image when $r < \sqrt{g+1}$. This note gives a candidate extension to the square endpoint $r^2 = g+1$. The new input is an equality-case refinement of their parabolic multiplication estimate. It extends their Artin vanishing from coefficient rank strictly below g through rank g . We then audit the resulting substitutions in their rigidity, integrality, nonabelian-Hodge, characteristic-extension, and socle arguments. This is a new, unrefereed candidate proof, not an established theorem.

Status and scope. The statement $g \geq r^2 - 1$ is **CANDIDATE**. The published strict range is due to Landesman–Litt. This note does not establish the sharper proposed range $g \geq r^2 - 4$, and it makes no new genus-three or genus-four rank-three claim. Successful compilation and finite arithmetic checks do not replace independent mathematical review.

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1 Statement and published inputs

Let $\Sigma_{g,n}$ be an orientable genus- g surface with n punctures. The mapping class group $\mathrm{Mod}_{g,n}$ acts by precomposition on conjugacy classes of representations of $\pi_1(\Sigma_{g,n})$. A representation is *MCG-finite* if its conjugacy class has finite orbit.

Theorem 1.1 (Square-endpoint finite-image theorem). *Let $g \geq 3$, $n \geq 0$, and $r^2 \leq g + 1$. Every MCG-finite representation*

$$\rho : \pi_1(\Sigma_{g,n}) \longrightarrow \mathrm{GL}_r(\mathbf{C})$$

has finite image.

The strict case $r^2 < g + 1$ is Landesman–Litt [1, Theorem 1.2.1]. Only $r^2 = g + 1$ is new here. The proof uses the following published statements in their cited forms.

- (I1) Landesman–Litt [1, Proposition 5.2.3]: if E_* is a semistable parabolic bundle of parabolic degree zero and rank d , and multiplication by a nonzero

$$v \in H^0(C, E \otimes \omega_C(D))$$

has rank s on $H^0(C, E^\vee \otimes \omega_C)$, then $d \geq g - s$.

- (I2) The proof of Landesman–Litt [2, Proposition 6.3.6], together with [2, Lemma 6.2.3]: for the coparabolic bundle used in [item \(I1\)](#), a generically generated subbundle A of corank c and section deficit δ gives $d \geq gc - \delta$. After quotienting the HN part N^t of slope greater than $2g - 2$, the same proof gives

$$\mathrm{rk}(F/N^t) \geq gc - \delta.$$

When $N^t = 0$, its complete inequality chain is

$$\deg F + (1 - g) \mathrm{rk} F \leq \frac{\deg A}{2} + \mathrm{rk} A + \delta \leq \frac{\mu(F) \mathrm{rk} A}{2} + \mathrm{rk} A + \delta. \quad (1)$$

- (I3) Landesman–Litt [1, Lemma 6.1.1 and Theorem 1.7.1]: if U is an irreducible unitary fibral local system and $0 \neq L \subset R^1\pi_*^\circ U$ is irreducible, then, after possible complex conjugation, a nonzero Hodge vector has multiplication rank at most $\mathrm{rk} L/2$; in rank one one has $\mathrm{rk} L \geq 2g - 2$.
- (I4) Landesman–Litt [1, Lemma 2.4.2 and Theorem 6.2.1]: after a dominant etale base change, a free Artin local system whose fibral restriction is a constant deformation of a unitary system decomposes into terms

$$U_i \otimes \pi^* W_i,$$

with U_i irreducible unitary on the fibres and W_i free over the Artin algebra. Their fixed-part bound proves vanishing when the total coefficient rank is strictly less than g .

- (I5) Landesman–Litt [1, Sections 8.2–8.5]: adjoint fixed-part vanishing gives strong cohomological rigidity and formal constancy. The remaining projective and linear integrality, unitary finiteness, and nonabelian-Hodge steps use no further occurrence of $r < \sqrt{g+1}$; the PVHS-to-unitary input is available in the larger range $r < 2\sqrt{g+1}$ [2, Theorem 1.2.13]. The canonical-representations paper cites the earlier numbering Theorem 1.2.12.
- (I6) Landesman–Litt [1, Theorem 7.2.1, Lemma 8.6.1, and Section 8.7]: for an irreducible finite-group representation σ , every nonzero finite-index-mapping-class-stable subspace of the relevant isotypic cohomology has dimension at least $2g - 2 \dim \sigma$. Comparing this with characteristic extension support and then inducting on the socle filtration proves the non-semisimple case in the published range.

For clarity about conventions, if E_* has canonical weights in $[0, 1)$, the parabolic bundle to which [item \(I2\)](#) is applied is

$$(E_*)^\vee \otimes \omega_C(D).$$

Its coparabolic degree-zero underlying ordinary bundle is $F = E^\vee \otimes \omega_C$. All ordinary section counts below concern this F .

2 The strict equality-case estimate

Proposition 2.1 (Strict stable pairing estimate). *Let C be a smooth projective curve of genus $g \geq 2$, let D be reduced, and let E_* be parabolically stable of parabolic degree zero and rank $d \geq 2$. For nonzero*

$$v \in H^0(C, E \otimes \omega_C(D)),$$

let s be the rank of the multiplication map

$$H^0(C, E^\vee \otimes \omega_C) \longrightarrow H^0(C, \omega_C^2(D)).$$

Then

$$\boxed{d \geq g - s + 1.}$$

Proof. The assertion is automatic when $s \geq g$, so suppose $s < g$. Input (I1) gives $d \geq g - s$. Assume for contradiction that equality holds:

$$d + s = g. \tag{2}$$

Put $F = E^\vee \otimes \omega_C$. Multiplication by v gives a nonzero map from F to the line bundle $\omega_C^2(D)$. Let K be its kernel. The image is torsion-free, so K is a saturated corank-one subbundle of F . Let $A \subset F$ be the saturation of the evaluation image of $H^0(C, K)$. Saturation stays in K because K is saturated. Every section of K factors through the evaluation image, while $A \subset K$, so

$$H^0(C, A) = H^0(C, K).$$

Consequently, with $c = d - \text{rk } A$ and $\delta = h^0(C, F) - h^0(C, A)$,

$$\delta = h^0(C, F) - h^0(C, K) = s.$$

Input (I2) now gives

$$d \geq gc - s.$$

Together with (2) and $c \geq 1$, this forces $c = 1$.

Let N^t be the sum of the HN pieces of A with slope greater than $2g - 2$. The proof of Proposition 6.3.6 preserves the deficit because $H^1(C, N^t) = 0$, and yields

$$\text{rk}(F/N^t) \geq gc - s = g - s = d.$$

Since F has rank d , it follows that $N^t = 0$.

The intermediate inequality in the proof of Lemma 6.2.3 is therefore

$$\mu(F) + 2 \leq (2g - \mu(F))d + 2s. \tag{3}$$

The coparabolic quotient-slope lemma gives $\mu(F) \geq 2g - 2$. At $\mu(F) = 2g - 2$, the two sides of (3) are equal by (2). Increasing $\mu(F)$ increases the left side and decreases the right side. Hence

$$\mu(F) = 2g - 2.$$

Write S for the total canonical parabolic weight of E_* . Since $\deg E = -S$,

$$\mu(F) = 2g - 2 + \frac{S}{d}.$$

Thus $S = 0$. The canonical weights are nonnegative, so every weight is zero. Because the weights in a parabolic flag are strictly increasing, each flag is the trivial one-step flag. Parabolic stability is

now ordinary stability of the degree-zero bundle E ; dualizing and twisting show that F is ordinary stable of slope $2g - 2$.

We may finally use the complete chain (1). Here $\text{rk } A = d - 1$, $\delta = s$, and $\mu(F) = 2g - 2$, so it reads

$$\deg F + (1 - g)d \leq \frac{\deg A}{2} + d - 1 + s \leq \frac{(2g - 2)(d - 1)}{2} + d - 1 + s.$$

The leftmost term is $d(g - 1)$, and the rightmost term is

$$g(d - 1) + s = d(g - 1)$$

by (2). Equality therefore holds throughout. In particular,

$$\deg A = (2g - 2)(d - 1), \quad \mu(A) = \mu(F).$$

But $d \geq 2$ makes A a nonzero proper subbundle of the stable bundle F , a contradiction. $\square$

Remark 2.2 (The rank-one exception). The condition $d \geq 2$ is necessary. If $D = \emptyset$ and $E \in \text{Pic}^0(C)$ is nontrivial, then $h^0(E\omega_C) = h^0(E^{-1}\omega_C) = g - 1$. Multiplication by a nonzero section of $E\omega_C$ is injective on $H^0(E^{-1}\omega_C)$, so $s = g - 1$ and the published weak bound is sharp.

3 Fixed parts and Artin vanishing through rank g

Corollary 3.1 (Strict fixed-part estimate). *Let U be an irreducible unitary fibral local system of rank d on a punctured versal genus- g family. Every nonzero irreducible sub-local system $L \subset R^1\pi_*U$ satisfies*

$$\text{rk } L \geq \begin{cases} 2g - 2, & d = 1, \\ 2g - 2d + 2, & d \geq 2. \end{cases}$$

Proof. For $d = 1$ this is input (I3). For $d \geq 2$, that input produces a nonzero Hodge vector whose multiplication rank s is at most $\text{rk } L/2$. Proposition 2.1 gives $s \geq g - d + 1$, hence $\text{rk } L \geq 2g - 2d + 2$. $\square$

Corollary 3.2 (Artin vanishing through rank g). *Let $g \geq 3$. In the setup of Landesman–Litt [1, Theorem 6.2.1], its conclusion remains valid when $\text{rk}_A \mathcal{V} \leq g$.*

Proof. Use input (I4). It suffices to treat a summand $U \otimes \pi^*W$, where $u = \text{rk } U$, $w = \text{rk } W$, and $uw \leq g$. The maximal-ideal filtration reduces to the complex residual coefficient W_0 . A nonzero invariant gives a nonzero morphism

$$W_0^\vee \longrightarrow R^1\pi_*^\circ U.$$

Choose an irreducible sub-local system in its nonzero image. Its rank is at most w , so Corollary 3.1 gives

$$w \geq \begin{cases} 2g - 2, & u = 1, \\ 2g - 2u + 2, & u \geq 2. \end{cases}$$

If $w = 1$, the displayed lower bound is at least two for every $u \leq g$, a contradiction. If $w \geq 2$ and $u = 1$, then $uw = w \geq 2g - 2 > g$. Finally, if $w \geq 2$ and $u \geq 2$, the constraint $uw \leq g$ gives $u \leq g/2$, while

$$uw \geq 2u(g - u + 1) \geq 2g > g.$$

Every case contradicts $uw \leq g$. $\square$

4 The square-endpoint pipeline

We now prove Theorem 1.1. The following table records the only substitutions in the published Landesman–Litt proof.

Published step	Strict-rank use	Endpoint replacement
Proposition 8.2.1	Theorem 6.2.1 for the rank- $r^2 - 1$ adjoint	Corollary 3.2; Schur and Leray are unchanged
Lemma 8.3.3	Strong rigidity from Proposition 8.2.1	Endpoint strong rigidity; no further rank estimate
Lemma 8.3.4	Projective integrality	Endpoint projective integrality; the finite-determinant lift is unchanged
Proposition 8.4.1	Rigidity, integrality, and the larger isomonodromy range	Endpoint rigidity and integrality; $r < 2\sqrt{g+1}$ is automatic
Lemma 8.5.1	Theorem 6.2.1 on every Artin truncation of the adjoint	Corollary 3.2
Lemma 8.5.2	Lemma 8.5.1	No additional numerical hypothesis
Lemma 8.6.1	Strict AM–GM bound on extension support	The endpoint comparison in Lemma 4.2
Section 8.7	Semisimple theorem and Lemma 8.6.1	The same socle induction

Proposition 4.1 (Semisimple square endpoint). *Assume $r^2 = g + 1$. Every semisimple MCG-finite rank- r representation has finite image.*

Proof. If the representation is reducible, each irreducible summand has rank strictly less than $r = \sqrt{g+1}$. Semisimple subquotients remain MCG-finite [1, Proposition 2.1.1], so the published strict theorem makes every summand finite.

Assume the representation is irreducible. Its trace-free adjoint has rank

$$r^2 - 1 = g.$$

Thus Corollary 3.2 replaces Theorem 6.2.1 in both strong cohomological rigidity [1, Proposition 8.2.1] and formal constancy [1, Lemma 8.5.1]. The projective and linear integrality arguments, unitary finiteness, and passage from formal to global constancy then apply as in [1, Sections 8.3–8.5]. The nonabelian-Hodge deformation ends at a unitary system because the required inequality is

$$r < 2\sqrt{g+1} = 2r.$$

The endpoint is therefore conjugate to a finite unitary representation. $\square$

Lemma 4.2 (Characteristic extensions at equality). *Assume $r^2 = g + 1$. Let an MCG-finite representation be an extension of two semisimple finite-image representations of positive ranks a, b with $a + b \leq r$, and suppose the subrepresentation is characteristic. Then the extension splits and has finite image.*

Proof. Decompose the finite-image coefficient as

$$\rho_2^\vee \otimes \rho_1 = \bigoplus_i \sigma_i^{\oplus n_i}, \quad d_i = \dim \sigma_i.$$

The AM–GM inequality gives

$$n_i d_i \leq ab \leq \frac{(a+b)^2}{4} \leq \frac{r^2}{4} = \frac{g+1}{4}.$$

In particular $d_i \leq (g+1)/4 < g$. After passing to the finite Galois cover that trivializes both factors, the construction in [1, Lemma 8.6.1] turns any nonzero projected extension into a finite-index-mapping-class-stable quotient of the relevant σ_i -isotypic homology component of dimension at most $n_i d_i$. Dualizing embeds the dual quotient as a stable subrepresentation of the dual $\sigma_i^\vee$ -isotypic cohomology component, with the same dimension; here $\dim \sigma_i^\vee = d_i$. Input (I6) therefore gives the lower bound

$$2g - 2d_i \geq 2g - \frac{g+1}{2} = \frac{3g-1}{2} > \frac{g+1}{4} \geq n_i d_i,$$

a contradiction. Hence every projected map vanishes on the finite cover. The representation is therefore trivial on a finite-index subgroup and has finite image. Over $\mathbf{C}$, Maschke semisimplicity then splits the extension. $\square$

Proof of Theorem 1.1. The strict case is published. At equality, the semisimple case is Proposition 4.1. For an arbitrary representation, apply the characteristic socle filtration exactly as in [1, Section 8.7]. Every proper factor has rank strictly less than $r = \sqrt{g+1}$ and is already finite by the published theorem; equivalently, one may induct on the length of the socle filtration. Lemma 4.2 splits each characteristic extension. The original representation therefore has finite image. $\square$

5 Audit boundary

The new mathematics in this note is concentrated in three places:

1. the equality case in Proposition 2.1, especially the deduction $N^t = 0$, the forced vanishing of every parabolic weight, and the final equality-of-slopes contradiction;
2. the rank- g endpoint in Corollary 3.2; and
3. the strict support margin in Lemma 4.2.

The rest is a source-level substitution into the published pipeline. This note accepts the cited Hodge, coparabolic, integrality, isomonodromy, and Putman–Wieland inputs in their published forms. It neither reproves those theorems nor claims that an internal reconstruction is a substitute for specialist review.

The first square pairs are

$$(g, r) = (3, 2), (8, 3), (15, 4), (24, 5), \dots$$

The proof does not use a Spin lift and therefore includes even rank at the square endpoint. It supplies no mechanism below $g = r^2 - 1$.

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References

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